

Contact

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www.linkedin.com/in/nour-karawani
(LinkedIn)

Top Skills

Python (Programming Language)

SQL

Jupyter Notebook

Languages

Arabic (Native or Bilingual)

Hebrew (Full Professional)

English (Full Professional)

Nour M. Karawani

Chemical Engineer | Process Engineering & R&D | Applied AI (ML & DL) | Industry 4.0 Systems | Data-Driven Processes

Israel

Summary

Driven by curiosity and a passion for problem-solving, I leverage my background in engineering, data analysis, and automation to optimize processes and drive innovation. Holding a B.Sc. in Chemical Engineering from BGU and a certification in Industry 4.0, I have practical experience in Python, SQL, MATLAB, and Advanced Excel for data-driven decision-making and process improvement.

At Atiko Labs, I developed & optimized signal processing, statistical data analysis, and pattern recognition algorithms to enhance Surface-Enhanced Raman Spectroscopy (SERS) analysis, enabling better detection of low-concentration molecules in complex media like blood. This work aims to revolutionize non-invasive early diagnostics, significantly impacting healthcare by improving the accuracy and efficiency of medical tests. Delivering real-time, on-site detection with trace-level sensitivity through a platform that is quick, accurate, accessible, and cost-effective.

At Moona, I developed an automated box-sorting system, integrating PLC programming, Yaskawa robotics, laser sensors, and Cognex cameras for precise and efficient sorting. By optimizing pneumatic motion control and RPA, I enhanced speed and adaptability to customer demands while reducing manual labor, minimizing energy consumption, and increasing operational efficiency. Additionally, I contributed to system design and prototyping with SolidWorks and 3D printing, improving sorting reliability and performance.

Currently expanding my expertise in AI and Data Science, I'm eager to apply my skills in process engineering, automation, and data analytics/science to tackle real-world challenges. I'm open to global opportunities where I can contribute to impactful projects and continue to grow in the tech industry.

Experience

Alrowad for Science and Technology
Engineering and Technology Instructor & Mentor
November 2025 - Present (5 months)

I teach and mentor students from elementary to high school in engineering and technology, creating engaging, hands-on learning experiences that spark curiosity and inspire innovation.

Atiko Labs
R&D Algorithm Developer & Data Analyst
June 2023 - November 2025 (2 years 6 months)

Developed a semi-automated algorithm for Surface-Enhanced Raman Spectroscopy (SERS) to detect low-concentration molecules in complex media, improving detection accuracy for early diagnostics and environmental monitoring.

Designed spectral peak detection methods, applied pattern recognition, and optimized signal-to-noise ratios for robust and reliable analysis.

Conducted data preprocessing, normalization, and dimensionality reduction using classical and AI-based techniques (PCA, PLS, PCR).

Contributed to peer-reviewed research and presented findings through documentation, oral presentation, and scientific poster. Enhanced the scalability and speed of spectral analysis, enabling practical applications in research and industrial settings.

Supervisor: Prof. Muhammad Bashouti.

Moona- A Space For Change
Industry 4.0 Trainee
November 2024 - February 2025 (4 months)

Completed hands-on Industry 4.0 training with a focus on industrial automation, robotics, and process optimization.

Developed PLC ladder diagrams, VisiLogic programs, and programmed Yaskawa robots for automated sorting systems. Designed 3D models and prototypes in SolidWorks, integrating sensors, Cognex cameras, and pneumatics for efficient motion control.

Built and optimized Arduino-based systems for smart traffic, gas detection, irrigation, and access control, incorporating real-time feedback and data visualization.

Gained practical experience in debugging, process automation, and sensor integration to improve operational efficiency.

טנגס

Chess Instructor

October 2022 - June 2024 (1 year 9 months)

Lead chess instruction, applying strategic thinking and problem-solving skills while motivating students and managing classroom dynamics.

The Jacob Blaustein Institutes for Desert Research (BIDR)

Water Treatment, Research Assistant

October 2022 - June 2023 (9 months)

Sde Boker, South District, Israel

Performed column experiments using zeolite and synthetic membrane experiments to efficiently remove ammonium from wastewater.

Supervisor: Prof. Oded Nir.

Education

ECOM SCHOOL

Certificate Program, AI Developer · (January 2025 - February 2026)

Ben-Gurion University of the Negev

Bachelor of Science - BS, Chemical Engineering, Energy, Water, and Advanced Technologies Track · (August 2024)